

Intro to Lagrange Mech WS

$$\mathcal{L} = T - V \text{ or } T - U$$

T : K.E. kinetic energy

U : P.E. potential energy

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} = Q_i$$

q_i : generalized coordinates (displacement in mechanical systems)

Q_i : generalized forces

↳ non-conservative forces in the system

Key: # of DOF = # of coordinates

= # of equations

For mechanical system, plug in $\mathcal{L} = T - U$

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{q}_i} - \underbrace{\frac{d}{dt} \frac{\partial U}{\partial \dot{q}_i}}_{=0} - \frac{\partial T}{\partial q_i} + \frac{\partial U}{\partial q_i} = Q_i$$

↳ = 0 as potential energy is not dependant on velocity!

$$\textcircled{1} \frac{d}{dt} \frac{\partial T}{\partial \dot{q}_i} - \textcircled{2} \frac{\partial T}{\partial q_i} + \textcircled{3} \frac{\partial U}{\partial q_i} = \textcircled{4} Q_i$$

↳ sth to do with 'work'

Set the Equation of Motion, i times

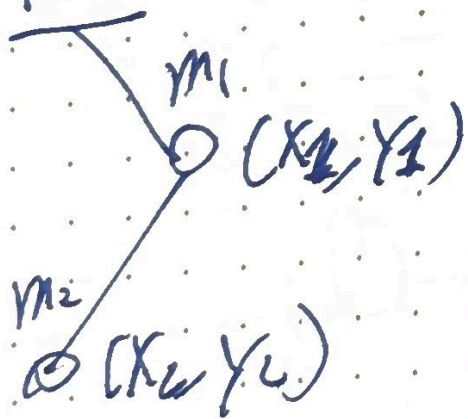
- What is this generalization about?
- 1) Don't need to be cartesian ($\hat{i}, \hat{j}, \hat{k}$ or $\hat{x}, \hat{y}, \hat{z}$)
 - 2) don't have to be ~~int~~ inertia frame (inertial)
 - 3) Must be independent, must be complete
 - 4) The system must be holonomic.

3.1) Independent: when you fix all but one coordinate, still have a continuous range of movement in the free coordinate

3.2) Complete: capable of locate all parts of the system at all times

assume $\dot{x}$ and x independent?

example: double pendulum



4 coordinates $(x_1, y_1), (x_2, y_2)$

2 DOF

NOT independent.

if we ~~fix~~ (x_1, y_1) ~~for the 2 DOF~~
then the system will be frozen

$L(x, \dot{x})$
mechanically
purely as
independent
variables

4) Holonomic: $\# \text{DOF} = \# \text{coordinates}$

'sphere'
Rolling Ball on a $x-y$ plane
is ~~not~~ non-holonomic system demo.

Here's the systematic way to "Lagrange"

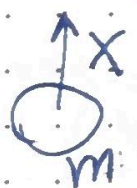
- ① Determine DOF, ~~choose~~ choose your q_i
- ② verify complete, independent holonomic
- ③ compute $L = T - U$
- ④ compute E-L equation for each q_i

LHS: $\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i}$

RHS: computing the virtual work δW associated with the virtual displacement δq_i

δW for non-conservative forces

example #1: $\delta W_i = Q_i \delta q_i$

 $T = \frac{1}{2} m \dot{x}^2$ $U = mgx$

$L = \frac{1}{2} m \dot{x}^2 - mgx$

$\frac{\partial L}{\partial \dot{x}} = m\dot{x}$ $\frac{\partial L}{\partial x} = -mg$

$\frac{d}{dt} \frac{\partial L}{\partial \dot{x}} = m\ddot{x}$

No friction,
No Q_i

$m\ddot{x} - (-mg) = 0$

$\ddot{x} = -g$

What is δ ? $= \Delta$
infinitesimal of $\sim$

δx very small, still has value

$2\delta x$ very small, still has value

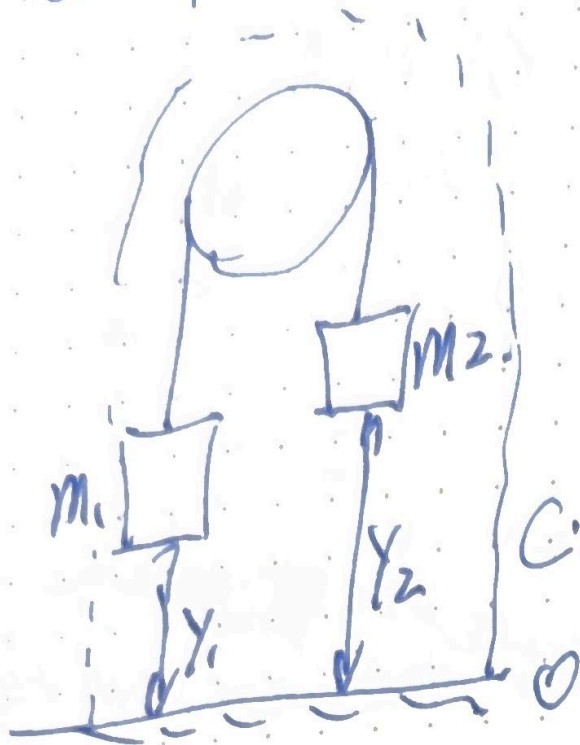
$\delta^2 x = 0$

that's why small angle
approx: $\sin \theta \approx \theta$
 $\cos \theta \approx 1$

$\sin \delta \theta = \delta \theta$

$\cos \delta \theta = 1$

example #2: ATWOOD MACHINE



C: total length

1 qd is enough!

But Let's do 2: y_1, y_2

$$T = \frac{1}{2} m_1 \dot{y}_1^2 + \frac{1}{2} m_2 \dot{y}_2^2$$

$$V = m_1 g y_1 + m_2 g y_2$$

$$\text{cauz: } y_1 + y_2 = C$$

$$y_2 = C - y_1$$

$$\dot{y}_2 = -\dot{y}_1$$

$$T = \frac{1}{2} m_1 \dot{y}_1^2 + \frac{1}{2} m_2 (-\dot{y}_1)^2 = \frac{1}{2} (m_1 + m_2) \dot{y}_1^2$$

$$V = m_1 g y_1 + m_2 g (C - y_1) = (m_1 - m_2) g y_1 + m_2 g C$$

$$\mathcal{L} = \frac{1}{2} (m_1 + m_2) \dot{y}_1^2 - (m_1 - m_2) g y_1 - m_2 g C$$

$$\frac{\partial \mathcal{L}}{\partial \dot{y}_1} = (m_1 + m_2) \dot{y}_1 \quad \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{y}_1} = (m_1 + m_2) \ddot{y}_1$$

$$\frac{\partial \mathcal{L}}{\partial y_1} = -(m_1 - m_2) g$$

$$\text{E-L: } (m_1 + m_2) \ddot{y}_1 + (m_1 - m_2) g = 0$$

$$\ddot{y}_1 = \frac{m_2 - m_1}{m_1 + m_2} g$$

$\mathcal{L}$ is good, as both T and V are

No friction
No Qi!

example #1 $\rightarrow$ simple pendulum

$$\text{DOF} = 2(x, y) - 1 \text{ (because of } L) = 1$$

coordinate, $q_1 = \theta$ $\dot{q}_1 = \dot{\theta}$

point of mass,
so no rotation

$$T = \frac{m}{2} (\dot{x}^2 + \dot{y}^2)$$

$$= \frac{m}{2} (\cos^2 \theta L^2 + \sin^2 \theta L^2) \dot{\theta}^2$$

$$= \frac{m}{2} L^2 \dot{\theta}^2$$

$$\sin \theta L = x$$

$$\cos \theta L = y$$

$$\dot{x} = L \cos \theta \dot{\theta}$$

$$\dot{y} = -L \sin \theta \dot{\theta}$$

$$V = +mgh = mg(L - L \cos \theta) = mgL(1 - \cos \theta)$$

$$\mathcal{L} = \frac{m}{2} L^2 \dot{\theta}^2 - mgL(1 - \cos \theta)$$

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = \frac{m}{2} L^2 \dot{\theta} \frac{\partial}{\partial \dot{\theta}} = m L^2 \dot{\theta}$$

$$\frac{\partial \mathcal{L}}{\partial \theta} = \frac{\partial}{\partial \theta} (mgL \cos \theta) = -mg \sin \theta$$

$$m L^2 \ddot{\theta} + mg \sin \theta L = 0$$

$$\ddot{\theta} = -\frac{g}{L} \sin \theta$$

example #4

point mass moving on a spring wire, no friction!



DOF = 2, (r, θ)

$q_1 = r$ $q_2 = \theta$

$\dot{q}_1 = \dot{r}$ $\dot{q}_2 = \dot{\theta}$

$T = \frac{1}{2} m v^2 = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\theta}^2)$

$V = 0$

$\mathcal{L} = \frac{1}{2} m \dot{r}^2 + \frac{1}{2} m r^2 \dot{\theta}^2$

$\frac{\partial \mathcal{L}}{\partial \dot{r}} = m \dot{r}$

$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{r}} = m \ddot{r}$

$\frac{\partial \mathcal{L}}{\partial r} = m \dot{\theta}^2 r$

$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = m r^2 \dot{\theta}$

$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = m r^2 \ddot{\theta}$

$\frac{\partial \mathcal{L}}{\partial \theta} = 0$

invariant

$m \ddot{r} - m \dot{\theta}^2 r = 0$ $\ddot{r} = \dot{\theta}^2 r$

$m r^2 \ddot{\theta} = 0 \rightarrow \ddot{\theta} = 0$ constant $\dot{\theta}$

say angular momentum is conserved.

Let's say $\dot{\theta} = \omega$

$\ddot{r} = \omega^2 r$ $\ddot{r} - \omega^2 r = 0$

$r = e^{\lambda t}$ $\dot{r} = \lambda e^{\lambda t}$ $\ddot{r} = \lambda^2 e^{\lambda t}$

$(\lambda^2 - \omega^2) e^{\lambda t} = 0$

$\lambda = \pm \omega$

$r(t) = C_1 e^{\omega t} + C_2 e^{-\omega t}$

cyclic coordinate
is not $\mathcal{L}$ dependent
cyclic coordinate
is conserved

guess or ansatz law happened

OR call $v_r = \dot{r}$
 $\dot{v}_r = \omega^2 r$

$\begin{bmatrix} \dot{r} \\ \dot{v}_r \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \omega^2 & 0 \end{bmatrix} \begin{bmatrix} r \\ v_r \end{bmatrix}$

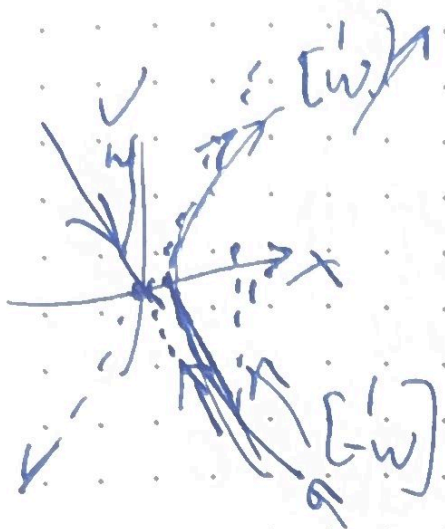
$\lambda = \pm i\omega \rightarrow$ unstable saddle

$\left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \dot{x} - \mathcal{L} \right)$
Hamilton
= E

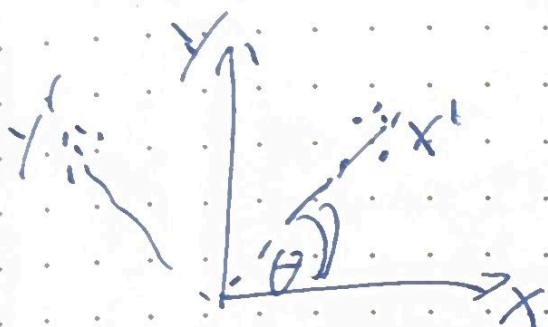
$$\begin{bmatrix} 0 & 1 \\ \omega^2 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \omega x \\ \omega y \end{bmatrix} \quad y = \omega x \quad \begin{bmatrix} 1 \\ \omega \end{bmatrix}$$

$$\omega^2 x = \omega x \quad \begin{bmatrix} 1 \\ -\omega \end{bmatrix}$$

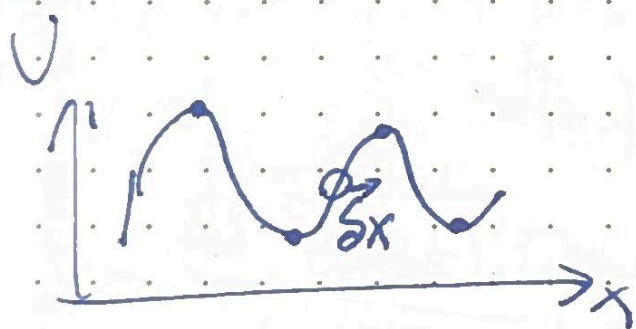
2.C. $\theta = 0, r = r_0$



Symmetry and conservation Laws.



Real forces derives from potential? conservation law
why Inertial frame? multiple points



$$\frac{\partial U}{\partial x} = -F$$

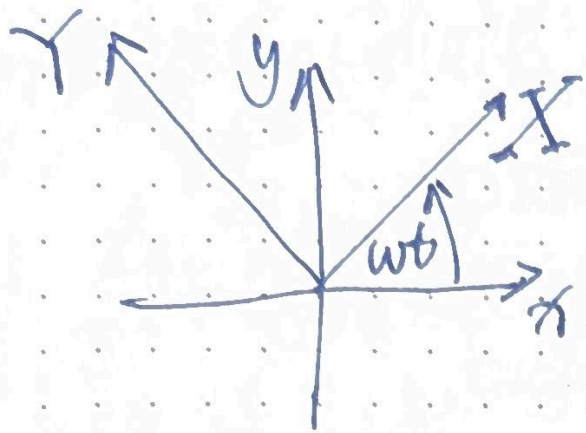
$$\frac{dU}{dx} \delta x = \delta U$$

rule: δx_m
equilibrium: $\delta U = 0$

potential doesn't change when moving a little bit of x (δx)

2.D? $\frac{\partial U}{\partial x} \delta x + \frac{\partial U}{\partial y} \delta y = \delta U$

equilibrium: $\delta U = 0$ for any possible $\delta x, \delta y$ (1st order of small distribution)
 (stationary points)



uniformly rotating coordinate system

$$x = X \cos \omega t + Y \sin \omega t$$

$$y = -X \sin \omega t + Y \cos \omega t$$

$\vec{X}, \vec{Y}$
is non-inertial frame

$$\mathcal{L}(x, y) = \frac{1}{2} m \dot{x}^2 + \frac{1}{2} m \dot{y}^2 \quad \text{no } V(x, y)$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}(x, y)}{\partial \dot{x}} = m \ddot{x}$$

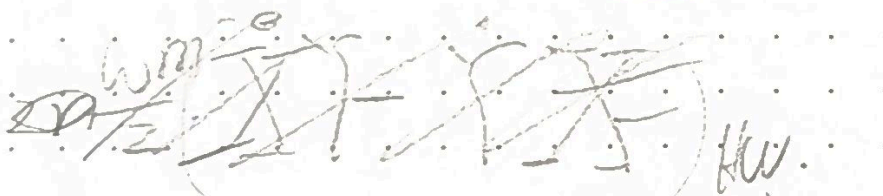
$$\frac{d}{dt} \frac{\partial \mathcal{L}(x, y)}{\partial \dot{y}} = m \ddot{y}$$

$$\dot{x} = \dot{X} \cos \omega t - \omega X \sin \omega t + \dot{Y} \sin \omega t + \omega Y \cos \omega t$$

$$\dot{y} = -\dot{X} \sin \omega t - \omega X \cos \omega t + \dot{Y} \cos \omega t - \omega Y \sin \omega t$$

$$\begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} = \begin{bmatrix} \cos \omega t & \sin \omega t \\ -\sin \omega t & \cos \omega t \end{bmatrix} \begin{bmatrix} \dot{X} \\ \dot{Y} \end{bmatrix} - \omega \begin{bmatrix} X \\ Y \end{bmatrix}$$

$$\mathcal{L}(\dot{X}, \dot{Y}) = \frac{m}{2} (\dot{X}^2 + \dot{Y}^2) + \frac{\omega^2 m}{2} (X^2 + Y^2) + m(\dot{X}Y - Y\dot{X})$$



$$\ddot{X} = -2\omega \dot{Y}$$

$$\ddot{Y} = 2\omega \dot{X}$$

looks like
potential energy

fictitious potential energy

$$U = -\frac{\omega^2 m}{2} (X^2 + Y^2)$$

centrifugal force

Coriolis
force

velocity
dependent
force

$$\frac{\partial \mathcal{L}(X, Y)}{\partial \dot{X}} = m \dot{X} + \omega m Y$$

$$\frac{\partial \mathcal{L}(X, Y)}{\partial \dot{Y}} = m \dot{Y} - \omega m X$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}(X, Y)}{\partial \dot{X}} = m \ddot{X} + \omega m \dot{Y}$$

$$m \ddot{X} + \omega m \dot{Y} - \omega m X + \omega m \dot{Y} = 0$$

$$m \ddot{X} = \omega m X - 2\omega m \dot{Y} \quad \text{Coriolis force}$$

$$F_{cor} = \begin{bmatrix} 2\omega m \dot{y} \\ -2\omega m \dot{x} \end{bmatrix}$$



~~$$F_{cor} = -cx$$~~

~~$$F_{cor} = -c$$~~

~~$$F_{cor} = 0$$~~

$$F_{cor} = \begin{bmatrix} 0 \\ -\omega m \dot{x} \end{bmatrix}$$

~~$$F_{cor} = -cx$$~~

~~$$F_{cor} = -c$$~~

$$F_{cor} = \begin{bmatrix} 0 \\ \omega m \dot{x} \end{bmatrix}$$